Transfer: circuits and induction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Apply circuit relationships and explain an induction observation.

SUCCESS CRITERIA - I can...

1. Use current and voltage rules in a new circuit.
2. Account for electrical power and energy.
3. Explain induction through changing flux.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A device receives 50 J mechanical energy at its shaft and produces electrical energy at its terminals. Identify it and name one possible loss.

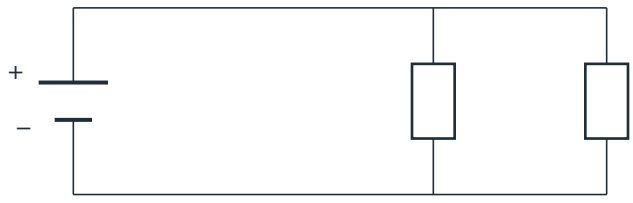
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $V = IR$; $P = IV$; $E = P\Delta t$; $|emf_{avg}| = N|\Delta\Phi|/\Delta t$. Use stated circuit or coil conditions. Ideal battery and wires unless stated otherwise.

Worked reasoning: A 12 V supply drives a 6 ohm resistor: current 2 A and power 24 W. A complete model checks both the charge-flow rate and the energy-transfer rate.

Your turn: Draw the circuit and an energy-flow arrow. Label what an ammeter and voltmeter would measure and where each connects.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Sketch two resistors in parallel across a battery and label the equal-voltage branches.



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Apply the physics

A 6 V battery powers 3 Ω and 6 Ω resistors in parallel. Label both branch voltages. Find each current, total source current and total power $P = VI_{total}$.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

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05 Explain your reasoning

A magnet is held motionless near a fixed coil, then moved into it. Explain the change in the meter response without saying the magnet supplies unlimited energy.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A 4 Ω resistor carries 6 A. Find the potential difference across it.
Draw the stated circuit and label quantities relevant to the question.

Draw the complete circuit and label shared nodes.

Marzano check - circle: ① ② ③

07 Exit check

A motor uses 6 W for 5 s. Find energy and name one possible non-mechanical output.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

08 STRETCH: test the model

Replace the resistor with two 12 ohm resistors in parallel. Compare total current and power, then draw both branch currents.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③